

*Answer all questions.*

1. The Energy Saving Trust aims to help everyone save energy every day in order to reduce the environmental damage that energy production does to the environment. It provides advice to households and businesses and gives tips for energy savings such as not leaving appliances on standby and choosing the most energy efficient appliances.

The efficiency of washing machines depends on how they are used as they can wash clothes at a variety of temperatures. Many clothes retailers and washing detergent manufacturers recommend washing clothes at 30 °C in order to be more energy efficient. According to the Energy Saving Trust “*Washing clothes at 30 °C rather than at higher temperatures uses around 40 % less energy.*”

The table below gives data for different washing temperatures for a particular washing machine.

| Washing temperature (°C) | Mean power (W) | Wash time (minutes) |
|--------------------------|----------------|---------------------|
| 30                       | 500            | 60                  |
| 40                       | 500            | 90                  |
| 50                       | 500            | 120                 |

- (a) (i) Use equations from page 2 to calculate the cost of washing clothes at 40 °C. The cost of one unit of electricity is 16 p. [4]

Cost = ..... p

- (ii) The Energy Saving Trust suggests that washing clothes at 30 °C reduces energy use by at least 40 % compared to washing at higher temperatures. Use the data in the table to investigate the validity of this claim. [3]

*Space for calculations*

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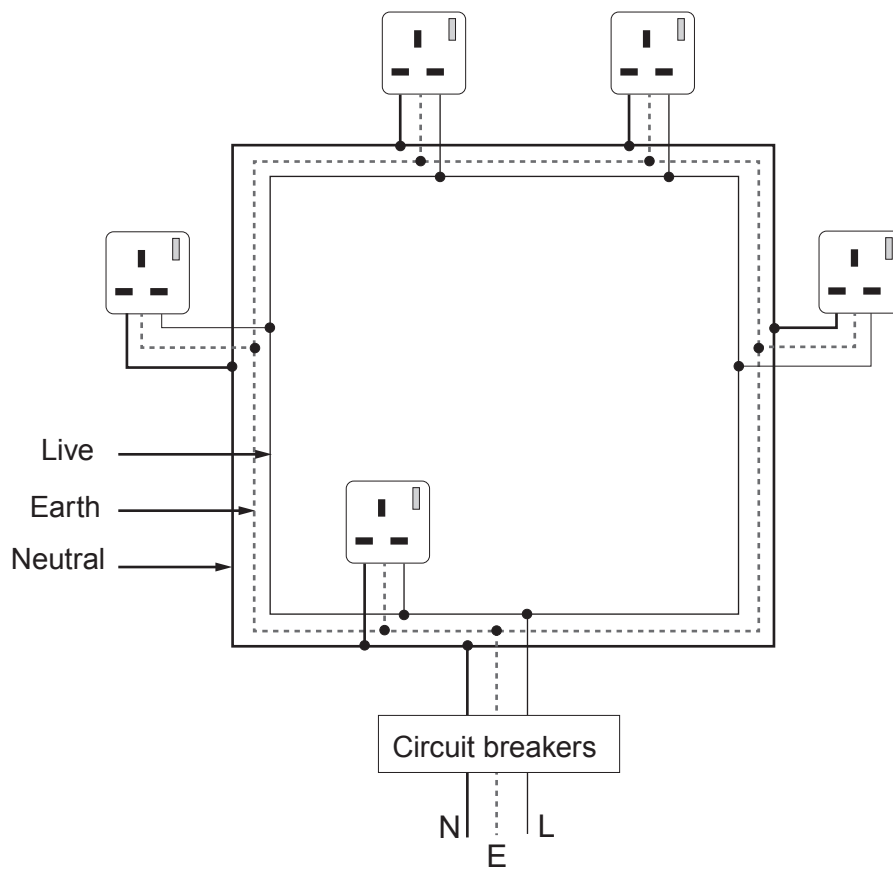
- (iii) Washing clothes at 30 °C saves money on household bills. Explain why reducing energy consumption also helps the environment. [2]

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- (b) The washing machine is connected to the ring main shown below. The voltage in the ring main is 230 V.



- (i) State which wire the fuse must be connected to. [1]

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- (ii) Describe the function of the earth wire. [2]

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- (c) When the voltage is 230V, the mean power of the washing machine during a wash is 500W. However when heating the water it must be able to produce a maximum power of 2400W. The householder has 3A, 5A and 13A fuses available and chooses to fit a 3A fuse into the plug. Investigate whether or not this is a suitable choice. [3]

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